

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1003	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:		Duration:	60 minutes

Code of Conduct

I _____ Keerthana.G (192521446) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

**AN AI-DRIVEN HOTEL RESERVATION AND ROOM MANAGEMENT PLATFORM WITH
DYNAMIC PRICING AND DEMAND FORECASTING**

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. Study and Analyze the Scenario

- Carefully understand the given problem statement.

The proposed system is an **AI-Driven Hotel Reservation and Room Management Platform** that automates the complete hotel reservation process while improving hotel operations using Artificial Intelligence (AI). Unlike traditional hotel management systems that use fixed room pricing and manual reservation methods, this system intelligently adjusts room prices based on customer demand, seasonal trends, occupancy rates, local events, and booking history. It also predicts future room demand using AI-based forecasting techniques.

The platform aims to simplify room booking for customers and provide hotel staff with tools to efficiently manage reservations, payments, housekeeping, room availability, and business reports. The AI component helps maximize hotel revenue by recommending the most suitable room prices and forecasting occupancy.

Detailed Analysis of the Problem Statement

1. The current hotel reservation process is often manual or partially automated.
2. Manual systems increase the chances of booking errors and overbooking.
3. Fixed room pricing does not reflect actual market demand.
4. Hotels struggle to estimate future occupancy accurately.

5. Managers cannot easily make pricing decisions based on market trends.
 6. Customers expect quick online booking and secure payment options.
 7. Hotels require a centralized platform to manage rooms, reservations, customers, housekeeping, and billing.
 8. AI can improve pricing decisions and demand prediction by analyzing historical booking data and customer behavior.
 9. The proposed system aims to automate hotel operations while increasing customer satisfaction and hotel profitability.
- Identify the business domain, stakeholders, existing challenges, and system objectives.

The proposed project belongs to the **Hospitality and Tourism Management** domain. It focuses on digital transformation of hotel operations using modern technologies such as Artificial Intelligence, Machine Learning, Cloud Computing, and Database Management Systems.

Business Area

- Hospitality Industry
- Hotel Reservation Services
- Tourism Management
- Online Booking Services

Technologies Used

- Artificial Intelligence (AI)
- Machine Learning (Demand Forecasting)
- Database Management System (MySQL)
- Web Technologies (HTML, CSS, JavaScript)
- Backend Technologies (Java, Python, PHP, etc.)
- Cloud Computing (optional)
- Online Payment Integration

Business Purpose

- Improve hotel operations.
- Increase hotel revenue.
- Reduce operational costs.
- Enhance customer satisfaction.
- Improve room occupancy.
- Automate pricing decisions.
- Generate business intelligence reports.

Business Benefits

- Faster booking process.
- Better customer experience.
- Improved room utilization.
- Reduced manual work.
- Accurate pricing strategies.
- Increased profitability.
- Better resource planning.

1.3 Identify Stakeholders

Stakeholders are individuals or organizations who use, manage, or benefit from the proposed system.

A. Primary Stakeholders

1. Customer (Guest)

Customers are the primary users of the system.

Responsibilities

- Register an account.
- Login securely.
- Search available rooms.

- Compare room types and prices.
- Book rooms.
- Make online payments.
- Modify reservations.
- Cancel reservations.
- View booking history.
- Submit reviews and ratings.

Benefits

- Faster booking.
- Real-time room availability.
- Secure payment.
- Fair AI-based pricing.
- Instant booking confirmation.

2. Receptionist

Receptionists manage daily hotel operations.

Responsibilities

- Check room availability.
- Create reservations.
- Modify reservations.
- Cancel bookings.
- Perform customer check-in.
- Perform customer check-out.
- Generate invoices.
- Handle customer queries.

Benefits

- Reduced paperwork.
- Faster reservation management.

- Automatic billing.
- Easy room allocation.

3. Hotel Manager

Managers supervise the overall hotel business.

Responsibilities

- Monitor room occupancy.
- View business reports.
- Approve pricing strategies.
- Monitor AI recommendations.
- Manage room inventory.
- Analyze revenue reports.
- Review demand forecasts.

Benefits

- Better decision making.
- Increased profitability.
- Efficient resource management.
- Business growth.

4. System Administrator

Administrators maintain the software.

Responsibilities

- Manage users.
- Create user accounts.
- Assign user roles.
- Configure system settings.
- Backup database.
- Monitor system security.

- Maintain software.

Benefits

- Centralized administration.
- Improved security.
- Easy maintenance.
- Reliable system operation.

5. Housekeeping Staff

Housekeeping staff manage room cleanliness.

Responsibilities

- View assigned rooms.
- Clean rooms.
- Update cleaning status.
- Notify reception after cleaning.

Benefits

- Better coordination.
- Faster room preparation.
- Real-time status updates.

B. Secondary Stakeholders

1. Hotel Owner

- Monitor hotel performance.
- Review profits.
- Track occupancy.
- Make investment decisions.

2. Payment Gateway Provider

- Process online payments.
- Verify payment transactions.
- Return payment status.
- Handle refunds.

3. AI Prediction Engine

- Analyze booking history.
- Predict occupancy.
- Recommend room prices.
- Forecast seasonal demand.

4. Email/SMS Service Provider

- Send booking confirmation.
- Send payment confirmation.
- Send cancellation alerts.
- Send promotional messages.

5. Third-Party Booking Platforms

- Share room availability.
- Receive booking requests.
- Synchronize reservations.

6. Government Tax Authorities

- Verify GST calculations.
- Ensure tax compliance.
- Maintain financial records.

1.4 Existing Challenges

Many hotels still rely on traditional reservation systems that have several limitations.

Challenge 1: Manual Reservation Process

- Employees manually record bookings.
- Increased paperwork.
- Slow booking process.
- Higher chances of mistakes.

Challenge 2: Overbooking

- Same room may be booked twice.
- Causes customer dissatisfaction.
- Results in financial loss.
- Damages hotel reputation.

Challenge 3: Fixed Room Pricing

- Room prices remain the same regardless of demand.
- Hotels lose revenue during peak seasons.
- Low occupancy during off-seasons.

Challenge 4: Poor Demand Prediction

- Hotels cannot accurately estimate future occupancy.
- Difficult to plan staffing.
- Difficult to manage room inventory.

Challenge 5: Lack of Real-Time Room Availability

- Customers may see outdated room availability.
- Delayed room updates.
- Booking conflicts occur.

Challenge 6: Inefficient Housekeeping Management

- Cleaning assignments are manual.
- Delays in room preparation.

- Rooms remain unavailable longer.

Challenge 7: Slow Payment Processing

- Manual billing.
- Delayed confirmation.
- Increased customer waiting time.

Challenge 8: Limited Reporting

- Managers receive only basic reports.
- Difficult to analyze revenue.
- Difficult to identify business trends.

Challenge 9: Poor Customer Experience

- Long booking process.
- Limited online services.
- Slow response to customer requests.

Challenge 10: High Operational Cost

- More employees required.
- Increased paperwork.
- Higher maintenance costs.

1.5 System Objectives

The proposed system aims to solve the above challenges by achieving the following objectives.

Primary Objectives

1. Automate the hotel reservation process.
2. Provide real-time room availability.
3. Implement AI-based dynamic pricing.

4. Forecast future room demand.
5. Reduce booking errors.
6. Eliminate overbooking.
7. Improve room utilization.
8. Increase hotel revenue.
9. Improve customer satisfaction.
10. Reduce manual paperwork.
11. Automate payment processing.
12. Improve housekeeping management.
13. Generate business reports.
14. Provide secure user authentication.
15. Maintain customer records.
16. Enable online booking from anywhere.
17. Support multiple user roles.
18. Improve operational efficiency.
19. Assist managers in business decision-making.
20. Build a scalable and intelligent hotel management platform.

Expected Outcomes

- Faster hotel reservation process.
- Improved customer satisfaction.
- Better occupancy management.
- Accurate AI-based room pricing.
- Increased hotel profits.
- Better resource utilization.
- Reduced operational costs.
- Secure and reliable hotel management.
- Data-driven business decisions.
- Overall improvement in hotel service quality.

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

Functional requirements describe the operations and services that the system should perform.

User Management

- The system should allow users to register and log in securely.
- Users should be able to update their profile and reset passwords.
- Different user roles such as Customer, Receptionist, Manager, Administrator, and Housekeeping Staff should be supported.

Room Management

- Add, update, and delete room details.
- Display real-time room availability.
- Categorize rooms based on type and facilities.
- Update room status (Available, Booked, Occupied, Cleaning, Maintenance).

Reservation Management

- Search rooms based on dates, room type, and price.
- Book, modify, and cancel reservations.
- Generate booking confirmation with a unique booking ID.
- Prevent duplicate bookings and overbooking.

Customer Management

- Store customer information securely.
- Maintain booking history.
- Allow customers to provide ratings and feedback.

Payment Management

- secure online payment through multiple payment methods.
- Generate invoices and payment receipts.
- Process refunds for cancelled bookings.

AI-Based Dynamic Pricing

- Analyze booking history and occupancy.
- Adjust room prices based on demand, season, and special events.
- Recommend optimal pricing to maximize revenue.

Demand Forecasting

- Predict future room occupancy using AI.
- Generate demand forecasting reports.
- Help managers plan staffing and room allocation.

Housekeeping Management

- Assign rooms to housekeeping staff.
- Update room cleaning status.
- Notify reception when rooms are ready.

Reports and Notifications

- Generate booking, occupancy, revenue, and customer reports.
- Send booking confirmations, payment alerts, cancellation notifications, and check-in reminders through email or SMS.

- Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

Performance

- The system should respond to requests within **3 seconds**.
- It should support multiple users simultaneously without performance issues.

Security

- User authentication and authorization should be implemented.
- Passwords should be encrypted.
- Customer and payment data should be protected using secure protocols.

Reliability

- The system should have **99.9% availability**.
- Regular backups should be maintained to prevent data loss.

Usability

- The interface should be simple, attractive, and easy to use.
- The booking process should be user-friendly for all users.

Scalability

- The system should support multiple hotels or branches.
- It should handle increasing numbers of users and bookings.

Maintainability

- The software should be modular for easy updates and maintenance.
- Errors should be easy to identify and fix.

Availability

- The system should be available **24×7** with minimal downtime.
- Users should be able to access the system from anywhere with an internet connection.

Portability

- The application should work on desktops, laptops, tablets, and smartphones.
- It should be compatible with major web browsers.

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

Primary Actors

1. Customer (Guest)

The customer uses the system to search for rooms, make reservations, complete payments, and manage bookings.

Use Cases

- Register/Login
- Search Rooms
- Book Room
- Make Payment
- Modify or Cancel Booking
- View Booking History
- Submit Feedback

2. Receptionist

The receptionist manages daily reservation activities and customer check-in/check-out.

Use Cases

- Check Room Availability
- Create or Modify Reservations
- Check-In Customer
- Check-Out Customer
- Generate Invoice

3. Hotel Manager

The manager monitors hotel operations and business performance.

Use Cases

- Manage Rooms
- View Occupancy Reports
- Monitor Revenue
- Review AI Pricing
- View Demand Forecast

4. System Administrator

The administrator manages users and maintains the system.

Use Cases

- Manage User Accounts
- Assign User Roles
- Configure System
- Backup Database
- Monitor Security

5. Housekeeping Staff

The housekeeping staff update room cleaning status.

Use Cases

- View Assigned Rooms
- Update Cleaning Status
- Mark Rooms as Ready

3.2 Secondary Actors

AI Prediction Engine

- Forecast room demand.
- Recommend dynamic room prices.

Payment Gateway

- Process online payments.
- Verify transactions.
- Handle refunds.

Email/SMS Notification Service

- Send booking confirmations.
- Send payment and cancellation notifications.
- Send check-in reminders.

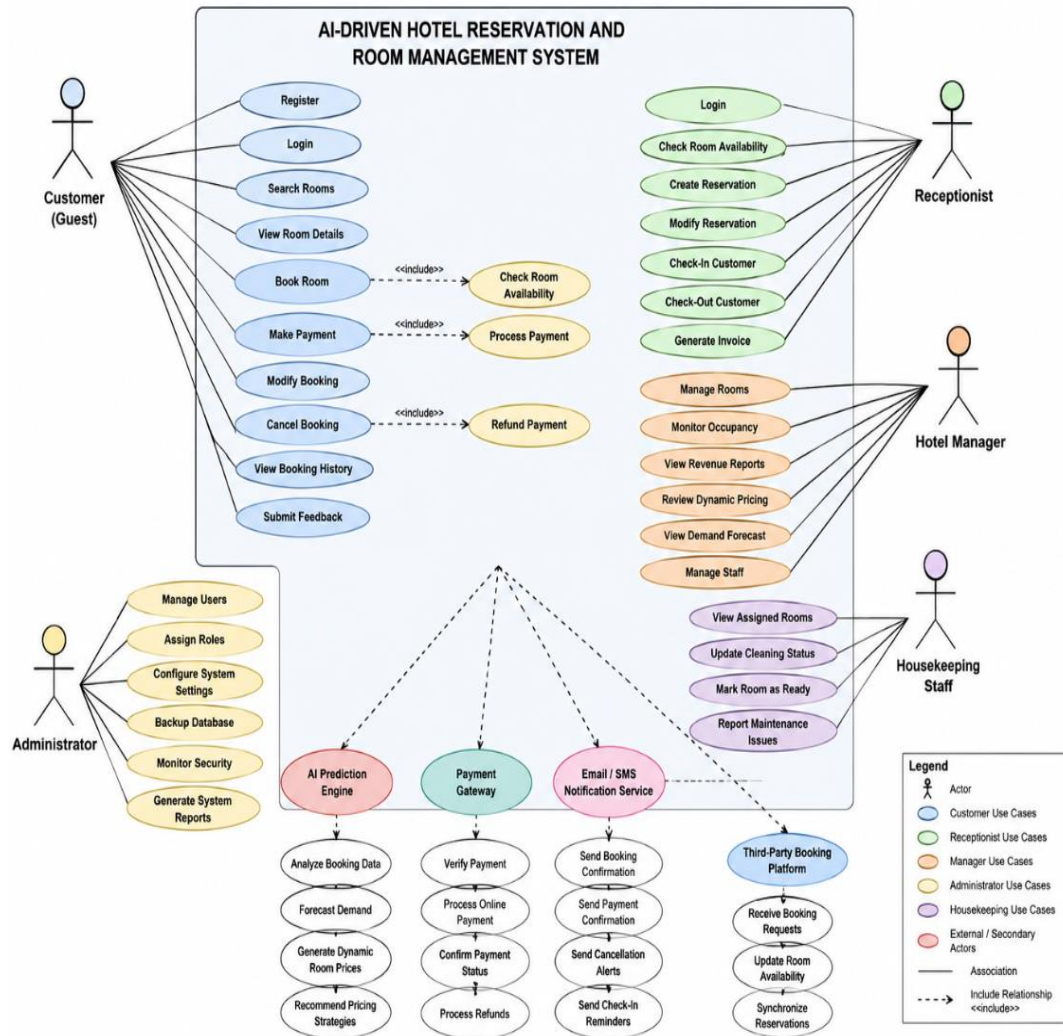
Third-Party Booking Platforms

- Share room availability.
- Synchronize reservations.

- List the major use cases associated with each actor.

Actor	Major Use Cases
Customer (Guest)	Register, Login, Search Rooms, View Room Details, Book Room, Make Payment, Modify Booking, Cancel Booking, View Booking History, Submit Feedback
Receptionist	Login, Check Room Availability, Create Reservation, Modify Reservation, Check-In Customer, Check-Out Customer, Generate Invoice
Hotel Manager	Manage Rooms, Monitor Occupancy, View Revenue Reports, Review Dynamic Pricing, View Demand Forecast, Manage Staff
System Administrator	Manage Users, Assign Roles, Configure System Settings, Backup Database, Monitor Security, Generate System Reports
Housekeeping Staff	View Assigned Rooms, Update Cleaning Status, Mark Room as Ready, Report Maintenance Issues
Payment Gateway	Verify Payment, Process Online Payment, Confirm Payment Status, Process Refunds

- Draw a Use Case Diagram representing the interactions between actors and the system.



4. Prepare the Software Requirement Specification (SRS)

Develop an SRS document containing:

- Introduction

The **AI-Driven Hotel Reservation and Room Management Platform** is a smart hotel management system designed to automate hotel operations using Artificial Intelligence

(AI). The system enables customers to search for available rooms, make reservations, complete online payments, and receive instant booking confirmations. It also helps hotel staff efficiently manage reservations, room availability, housekeeping, and customer information.

A key feature of the system is **AI-based Dynamic Pricing**, which automatically adjusts room prices based on demand, occupancy, seasonal trends, holidays, and historical booking data. Another important feature is **Demand Forecasting**, where AI predicts future occupancy levels to help hotel managers make better business decisions. Overall, the system improves operational efficiency, enhances customer satisfaction, and increases hotel revenue.

- Problem Description

Many hotels still use traditional reservation systems that rely on manual processes and fixed room pricing. These methods often lead to booking errors, overbooking, delayed room updates, inefficient resource management, and revenue loss.

Some of the major problems faced by hotels include:

- Manual reservation process increases the chances of human errors.
- Fixed room pricing does not reflect market demand.
- Overbooking due to inaccurate room availability updates.
- Difficulty in predicting customer demand during holidays and peak seasons.
- Poor coordination between reservation and housekeeping departments.
- Delayed payment processing and billing.
- Lack of analytical reports for decision-making.
- Low customer satisfaction due to slow booking services.

The proposed AI-based system addresses these challenges by automating hotel operations and using AI to optimize pricing and forecast demand.

- Scope of the System

The scope of the system defines the functionalities and services provided by the application.

The system will:

- Allow customers to register and log in securely.
- Display real-time room availability.
- Enable online room booking and reservation management.
- Support secure online payment methods.
- Automatically adjust room prices using AI.
- Forecast future room demand using historical data.
- Manage room allocation and housekeeping activities.
- Generate reports on occupancy, revenue, bookings, and customer details.
- Send booking confirmations and notifications through email or SMS.
- Provide separate access for customers, receptionists, managers, administrators, and housekeeping staff.
- Support future expansion to multiple hotel branches.

- Functional Requirements

Functional requirements describe the services that the system must provide.

4.1 User Management

- Users can register with personal details.
- Registered users can log in using secure credentials.
- Password reset and profile update options are available.
- The system supports role-based access for different users.

4.2 Room Management

- Add, update, and delete room information.
- Display room availability in real time.
- Categorize rooms based on type and facilities.
- Update room status (Available, Occupied, Cleaning, Maintenance).

4.3 Reservation Management

- Search rooms based on check-in/check-out dates.
- Book available rooms online.
- Modify or cancel existing reservations.
- Generate booking confirmation with a unique booking ID.
- Prevent duplicate bookings and overbooking.

4.4 Customer Management

- Store customer details securely.
- Maintain booking history.
- Collect customer feedback and ratings.

4.5 Payment Management

- Process online payments securely.
- Support multiple payment methods.
- Generate invoices and receipts.
- Process refunds for cancelled bookings.

4.6 AI Dynamic Pricing

- Analyze occupancy rates and booking history.
- Adjust room prices automatically.
- Recommend optimal pricing strategies.

4.7 Demand Forecasting

- Predict future occupancy levels.
- Forecast seasonal demand.
- Generate demand analysis reports.

4.8 Housekeeping Management

- Assign cleaning tasks.
- Update cleaning status.
- Notify reception when rooms are ready.

4.9 Reporting

- Generate occupancy reports.
- Generate revenue reports.
- Generate booking reports.
- Generate forecasting reports.

4.10 Notification System

- Send booking confirmations.
- Send payment confirmations.
- Send cancellation alerts.
- Send check-in reminders.

○ Non-Functional Requirements

Non-functional requirements describe the quality and performance standards of the system.

Performance

- System response time should be less than **3 seconds**.
- Support at least **1000 concurrent users**.

- Process bookings and payments quickly.

Security

- Secure login using authentication.
- Encrypt passwords and sensitive customer data.
- Protect payment information using secure protocols.
- Implement role-based access control.

Reliability

- Maintain **99.9% system uptime**.
- Perform automatic data backups.
- Recover data in case of system failures.

Usability

- Provide a simple and user-friendly interface.
- Ensure easy navigation.
- Make the booking process quick and simple.

- Use Case Descriptions

Use Case 1 – Book Room

Actor: Customer

Description: Allows customers to reserve available hotel rooms.

Preconditions

- Customer must be registered and logged in.
- Rooms must be available.

Main Flow

1. Customer searches available rooms.
2. System displays available rooms with prices.
3. Customer selects a room.
4. AI calculates the room price.
5. Customer confirms the booking.
6. Customer completes online payment.
7. System generates booking confirmation.

Postconditions

- Reservation is stored in the database.
- Room availability is updated.

Use Case 2 – Dynamic Pricing

Actor: AI Prediction Engine

Description: Automatically determines room prices based on demand and occupancy.

Preconditions

- Historical booking data is available.

Main Flow

1. Analyze booking history.
2. Check occupancy rate.
3. Identify seasonal demand.
4. Calculate optimal room price.
5. Update room pricing.

Postconditions

- Updated room prices are displayed.

- Data Requirements

The system stores and manages various types of data.

Customer Data

- Customer ID
- Name
- Email
- Phone Number
- Address
- Login Credentials

Room Data

- Room ID
- Room Number
- Room Type
- Capacity
- Price
- Availability Status

Reservation Data

- Reservation ID
- Customer ID
- Room ID
- Check-In Date
- Check-Out Date
- Booking Status

Payment Data

- Payment ID

- Reservation ID
- Payment Amount
- Payment Method
- Transaction Status

Housekeeping Data

- Staff ID
- Room ID
- Cleaning Status
- Assigned Date

Forecast Data

- Forecast ID
- Date
- Predicted Occupancy
- Suggested Room Price

- External Interface Requirements

User Interface

- Responsive web-based interface.
- Separate dashboards for customers, receptionists, managers, administrators, and housekeeping staff.
- Mobile-friendly design.

Hardware Interface

- Desktop computers
- Laptops
- Tablets
- Smartphones

- Printer (for invoices)

Software Interface

- MySQL Database
- AI Prediction Module
- Payment Gateway APIs
- Email and SMS Services
- Web Browser

Communication Interface

- Internet connection
- HTTPS protocol
- REST APIs
- Email and SMS integration

○ System Features

The AI-Driven Hotel Reservation and Room Management Platform includes the following features:

- User Registration and Secure Login
- Role-Based Access Control
- Online Room Search and Reservation
- Real-Time Room Availability
- AI-Based Dynamic Pricing
- AI Demand Forecasting
- Secure Online Payment
- Booking Confirmation and Notifications
- Customer Profile Management
- Housekeeping Management
- Check-In and Check-Out Management

- Revenue and Occupancy Reports
- Business Analytics Dashboard
- Customer Feedback Management
- Multi-Branch Hotel Support
- Secure Database with Automatic Backup
- Responsive Web Interface
- Integration with Payment Gateway and Email/SMS Services

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.

Assumptions are the conditions considered to be true while designing and developing the system.

1. Users have a stable internet connection to access the hotel reservation platform.
2. Customers provide accurate personal and booking information during registration and reservation.
3. Hotel staff regularly update room availability, housekeeping status, and booking details.
4. Historical booking data is available for the AI module to perform dynamic pricing and demand forecasting.
5. The payment gateway functions properly for secure online transactions.
6. The AI model provides reasonably accurate predictions based on available data.
7. The hotel has the necessary hardware and software infrastructure to operate the system.
8. All users have valid login credentials and appropriate access permissions.
9. The database is maintained regularly to ensure data accuracy and consistency.
10. The system will be used according to the hotel's operational policies and procedures.

- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Constraints are the limitations or restrictions that affect the design, development, and implementation of the system.

A. Technical Constraints

- The system requires a stable internet connection for online booking and payment processing.
- AI-based forecasting depends on the availability and quality of historical booking data.
- Integration with external services such as payment gateways and email/SMS APIs is required.
- The system must be compatible with different browsers and operating systems.
- Database performance may affect system response during peak usage.

B. Operational Constraints

- Hotel staff must be trained to use the system effectively.
- Room status and reservation details should be updated regularly to maintain data accuracy.
- Periodic maintenance is required to ensure smooth system operation.
- System performance depends on proper coordination between different hotel departments.

C. Economic Constraints

- Budget limitations may affect the selection of technologies and AI tools.
- Costs are involved in software development, testing, deployment, and maintenance.
- Additional expenses may arise from cloud hosting, payment gateway integration, and third-party services.
- Regular maintenance and system upgrades require financial resources.

D. Legal Constraints

- Customer personal information must be protected according to data privacy regulations.
- Online payment processing must comply with secure payment standards.
- The system should maintain accurate financial records for taxation and auditing purposes.
- Software licenses and third-party APIs must be used according to their legal agreements.

E. Time-Related Constraints

- The project must be completed within the planned development schedule.
- Testing and bug fixing should be completed before system deployment.
- AI model training and validation must be finished before implementation.
- Future updates and enhancements should be delivered within the planned maintenance timeline.

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.

The proposed system has been analyzed to ensure that the needs of all stakeholders are included.

Customer (Guest)

- User registration and secure login.
- Search and view available rooms.
- Online room booking and cancellation.

- Secure online payment.
- Booking confirmation and notifications.
- View booking history and submit feedback.

Receptionist

- Manage reservations.
- Check room availability.
- Perform customer check-in and check-out.
- Generate invoices and update booking details.

Hotel Manager

- Manage room details and availability.
- Monitor occupancy and revenue reports.
- View AI-based dynamic pricing suggestions.
- Analyze demand forecasting reports.

System Administrator

- Manage user accounts and roles.
- Configure system settings.
- Maintain database backup and security.
- Monitor overall system performance.

Housekeeping Staff

- View assigned rooms.
- Update room cleaning status.
- Mark rooms as ready for new bookings.

External Services

- Payment Gateway processes secure online transactions.
- AI Engine performs dynamic pricing and demand forecasting.
- Email/SMS Service sends booking confirmations and notifications.

Validation

All major stakeholder requirements have been identified and included in the proposed system.

- Check for ambiguity, redundancy, inconsistency, and missing requirements.

A. Ambiguity Check

- Requirements are written in clear and simple language.
- Functional requirements clearly describe the expected system behavior.
- Non-functional requirements specify measurable values such as response time and system availability.

Example:

"The system shall process room booking requests within **3 seconds**."

B. Redundancy Check

- Similar requirements are grouped together to avoid repetition.
- Duplicate functionalities have been removed.
- Each requirement is listed only once.

C. Consistency Check

- Room availability is updated automatically after booking or cancellation.
- Dynamic pricing is consistent with occupancy and demand forecasting.
- Payment confirmation is linked to successful reservation.
- Housekeeping updates are reflected in room availability.

D. Missing Requirement Check

The SRS includes all essential modules such as:

- User Management

- Room Management
 - Reservation Management
 - Payment Processing
 - AI-Based Dynamic Pricing
 - Demand Forecasting
 - Housekeeping Management
 - Notification System
 - Reporting and Analytics
 - Security and Backup
- Suggest improvements to enhance the quality and completeness of the requirement specification.

The following improvements can enhance the quality and completeness of the SRS:

1. Add **multilingual support** for international customers.
2. Integrate an **AI chatbot** to provide 24×7 customer assistance.
3. Introduce a **customer loyalty and reward program**.
4. Support **multiple hotel branches** under a single platform.
5. Integrate with **online travel agencies (OTAs)** such as Booking.com and Expedia.
6. Add a **mobile application** for Android and iOS users.
7. Include **real-time business dashboards** with occupancy, revenue, and pricing analytics.
8. Implement **two-factor authentication (2FA)** for enhanced security.
9. Add **predictive maintenance alerts** for hotel facilities using AI.
10. Conduct regular requirement reviews with stakeholders to ensure the system continues to meet business needs.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.

Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.
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Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

